YANTRIKA Robotics & Coding Club, IACS Kolkata

Contributing to Independent Projects

A complete guide to sharing your own simulations and interactive models with the Yantrika community, from first idea to a listed project page.

SIMULATE | MODEL | EXPLORE | SHARE

Who is this for? Students, alumni, hobbyists and first-timers from any school, college or university. You do not need to be a Yantrika member, and you do not need to belong to an institution: independent contributors are very welcome.

What do I send? A simulation or interactive model related to science, technology, engineering or mathematics, shared as a link: a code repository, a website you host yourself, or a Drive folder with downloadable files.

What do I get? Once reviewed, your project is listed on the Yantrika website under **Independent Projects** as a hyperlink with your name and institution (if any). You keep ownership of your work.

Website: `yantrika-iacs.github.io`
Contact: contact@yantrika.in

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1. The big picture

The Independent Projects category is a showcase for work that students and curious people build on their own. If you have made a simulation, a visualiser or an interactive model of something in STEM, you can submit it to be reviewed and listed on the Yantrika website, where visitors can find it, open it and learn from it.

Unlike the Games Arcade, your project is **not** added to our code. It stays where you built it (your repository, your website or your Drive), and the Yantrika site simply links to it and credits you. That means you choose the tools, the language and the style.

What you can do

- **Submit a project.** Fill in the submission form with your name, institution and a link.
- **Get reviewed.** A volunteer reviewer opens your link, checks it against the guidelines and may reply with suggestions.
- **Get listed.** Accepted projects appear on the Independent Projects page with your name, institution and a hyperlink.
- **Update later.** Improved your project? Write to us and we will refresh the listing.

Design principles

Principle	What it means for you
Open to everyone	Any student or user can submit. Independent contributors are as welcome as people from a college or school.
Your project, your home	We link to your work instead of copying it. You keep the code, the files and the copyright.
Any tool you like	Python, JavaScript, MATLAB, Julia, Scratch, a spreadsheet: if it simulates or models something well, it can be listed.
Reviewed before listing	Every submission is looked at by a person first, so visitors can trust what they click on.
Credit where due	Your name and institution are shown next to your project link.

2. What counts as an independent project

We are looking for **simulations** and **interactive models** with a clear link to a STEM subject. A visitor should be able to run it, play with it or step through it and come away understanding something a little better.

Type	What it is	Examples
Simulation	A program that imitates how a real system behaves over time.	Projectile motion with air drag, a random walk, heat flow in a rod, predator and prey populations.
Interactive model	A model with controls (sliders, buttons, inputs) that shows how results change.	Ohm's law explorer, Fourier series builder, lens ray tracer, a sorting algorithm visualiser.
Visualisation	A tool that turns data or a concept into a picture you can explore.	A graph theory visualiser, a phase portrait plotter, an orbit viewer.
Downloadable toolkit	A set of notebooks or files shared through Drive for others to run locally.	A Jupyter notebook series on numerical methods, with sample data.

Subjects we welcome

Physics, mathematics, chemistry, biology, computer science, electronics, robotics, statistics, data science, earth and environmental science, engineering and game theory. If your idea sits between two subjects, that is often where the best projects are.

What does not fit

- Projects with no STEM link, such as pure entertainment or marketing pages.
- Anything that needs visitors to create an account, pay, or hand over personal details.
- Plain slide decks, essays or videos with no simulation or interactive part.
- Work you did not make and do not have the right to share.

3. How to submit

Submission is a single online form. You give us the details below and one link to your project, then we take it from there.

The submission form

Field	Required	What to enter
Full name	Yes	The name you want shown next to your project.
Affiliated institution	Yes	Your school, college, university or organisation. If you do not belong to one, write **Independent contributor**.
Project title	Yes	A short, clear name, ideally under 50 characters.
STEM area	Yes	The main subject, for example Physics or Computer Science.
Short description	Yes	Two or three sentences: what it simulates or models, and what a visitor can do with it.
Project link	Yes	One link of the kind described below.
Link type	Yes	Repository, hosted website, or Drive media.
Contact email	Yes	So a reviewer can reach you with questions. It is never shown publicly.
Declaration	Yes	A tick-box confirming the project is your own work, or credited, and that you are happy for us to link to it.

Three ways to share your project

Link type	What to share	What we check
Repository	A link to a public repository on GitHub, GitLab, Codeberg or similar.	There is a README that explains what the project is and how to run it.
Hosted website	A link to a page you host yourself, for example on GitHub Pages, Netlify, Vercel or your own domain.	It opens without a login, loads reliably and works on a phone-sized screen.
Drive media	A Drive (or similar) link to files that can be downloaded: code, notebooks, executables, data and a short guide.	Sharing is set to "anyone with the link", files can be downloaded and a README is included.

Tip: not sure which to pick? If your project runs in a browser, a hosted website is the easiest for visitors. If it is Python or another language that needs installing, share a repository. Use Drive for notebooks, large data or compiled files.

4. What happens after you submit

Every submission follows the same simple path.

Step	What happens
1. Submitted	Your form arrives with the team. You do not need an account.
2. Review	A reviewer opens your link, tries the project and checks it against the review checklist in section 5.
3. Feedback	If something needs fixing, a reviewer emails you with specific, friendly suggestions. You can update and reply. This is normal and not a rejection.
4. Listed	Once approved, your project is added to the Independent Projects page as a hyperlink with your name and institution.
5. Kept current	Links are checked from time to time. If a link stops working, we will contact you before removing it.

How your project appears

Each listing is a hyperlink to your project, shown with your name and institution. For example:

> **Projectile Motion Lab**
> by Your Name, Your Institution (or Independent contributor)

Visitors click the title and go straight to your project. We do not host, copy or edit your files.

5. Review checklist

Reviewers look at the following. You can use this list to check your own work before you submit.

Area	Questions the reviewer asks
It works	Does the link open? Can the project be run or used start to finish without errors?
It is STEM	Is it clearly a simulation or interactive model related to science, technology, engineering or maths?
It is accurate	Is the science or maths correct, and is any simplification explained?
It is clear	Is there a short description, instructions on how to use or run it, and a note on what it teaches?
It is safe	No malware, hidden downloads, tracking, ads, pop-ups or requests for personal data. Content is suitable for school students.
It is yours	Is it your own work, or are the sources and licences of any third-party material credited?
It stays up	Is the link public and stable, not private, expiring or dependent on one login?

6. Preparing each kind of link

> **Check before you send:** Open your own link while signed out. If you can open and use it, so can a reviewer.

A repository

- Make the repository **public** before you submit. Private links cannot be reviewed.
- Add a README with a title, a one-paragraph summary, a screenshot or GIF if possible, and run instructions.
- List what is needed: language version, libraries, and the exact commands.
- Add a licence file so others know what they may do with your code.

```
# a README should let a stranger run your project
1. git clone https://github.com//
2. pip install -r requirements.txt
3. python main.py
```

A hosted website

- Test the link in a private or incognito window to make sure no login is needed.
- Check it on a narrow window and on a phone. Buttons and sliders should be easy to tap.
- Show a short "how to use this" note and, if you can, the theory behind the model.
- Avoid popups, ads, cookie walls and trackers.

Drive media

- Set sharing to **Anyone with the link can view** and make sure the files can be downloaded.
- Put everything in one folder, or one zip file, and add a README or a short PDF guide.
- Name files clearly and keep the size reasonable so that people on slow connections can get it.
- If a file needs special software, say which one and which version.

7. Rules every submission follows

#	Rule	Why
1	**Original or credited.** Submit only what you made or have the right to share. Credit any third-party code, images or data.	Keeps the project legally clean.
2	**Educational and kind.** The project should teach or celebrate STEM and be suitable for school students.	This is a learning space.
3	**No harmful content.** No malware, scams, phishing, hidden downloads or anything that damages a visitor's device.	Visitors must be safe.
4	**No personal data collection.** No required sign-ins, hidden tracking or collection of visitors' details.	Many visitors are school students.
5	**Truthful details.** Your name, institution and description must be accurate. Independent contributors should say so.	Credit should be honest.
6	**Keep the link alive.** Tell us if the link changes or the project moves.	Broken links frustrate learners.
7	**AI tools and licences.** If you used an AI tool, make sure its terms allow you to share the output, and say so in your README.	Avoids licensing surprises.

8. Licence, credit and ownership

- **You keep your copyright.** Yantrika only links to your project; we do not copy or relicense it.
- By submitting you give the club permission to **list your name, institution, project title and link** on the website.
- You decide the licence of your own project (MIT, GPL, CC BY and so on). Please state it in your repository or README.
- Please do not use Yantrika or IACS logos in a way that suggests official endorsement of your project.

Using someone else's material

Libraries, images, sounds, datasets and snippets are fine if their licence allows reuse. Keep the required notices and mention them in your README. Do not submit anything copied from a book, a website or another project whose terms do not allow reuse. When in doubt, ask us or leave it out.

9. Tips for a great project

- **Start with one idea.** A small model that is explained well beats a large one that is confusing.
- **Let people play.** Sliders, buttons and instant feedback make a simulation memorable.
- **Explain the science.** A short note on the equations or assumptions turns a toy into a teaching tool.
- **Show what it does.** A screenshot, GIF or short walkthrough in your README helps reviewers and visitors.
- **Be honest about limits.** Say what your model ignores, for example friction or quantum effects.
- **Make it accessible.** Use readable colours, labels on controls, and do not rely on colour alone.
- **Test with a friend.** Ask someone who has not seen it to try it and tell you where they got stuck.

Need an idea?

Choose something you studied this semester and make it move. Good starting points include pendulums and oscillators, Fourier series, ray optics, heat diffusion, random walks, cellular automata, graph algorithms, population dynamics, gas laws and circuit behaviour.

10. Troubleshooting

Problem	Likely cause and fix
The reviewer cannot open my link	The repository or folder is private, or Drive sharing is restricted. Set it to public or "anyone with the link" and test it signed out.
My website shows a blank page	Check the browser console (F12) for errors, and that file paths are correct and case-sensitive on hosts like GitHub Pages.
I do not belong to an institution	That is fine. Enter **Independent contributor** in the institution field.
I want to change my listing	Email us with the project title and what to update, such as a new link or a corrected name.
I want my project removed	Email us and we will take it down from the listing.
I have not heard back	Reviews are done by volunteers, so please be patient. If it has been a long while, send a polite reminder by email.
I am stuck	Write to us. Beginners' questions are welcome.

11. Cheat sheet

Quick summary

Question	Answer
Who can submit?	Any student or user, with or without an institution.
What can I submit?	Simulations or interactive models related to STEM.
How do I share it?	A repository link, a hosted website link, or a Drive link to downloadable media.
What is required?	Your name and your institution (or Independent contributor), plus the project details and link.
What happens next?	Review, then listing as a hyperlink with your name and institution.
Where do I ask questions?	contact@yantrika.in

Before you submit

- My project is a simulation or interactive model related to STEM
- My link is public and opens when I am signed out
- There is a README or short guide that says what it is and how to use it
- It is my own work, or any third-party material is credited
- It has no malware, tracking, ads or required sign-ins
- My name and institution (or "Independent contributor") are ready for the form

Thank you for sharing your work with us. Every project you build helps the next student learn something new. For any queries, write to **contact@yantrika.in**.